The Association Between Moderate Physical Activity and Self-Reported Depression Outcomes: Evidence from the National Health Interview Survey, 2010–2023

VANDERBILT UNIVERSITY

Nicolas Greco

Departments of Economics & Psychology, Vanderbilt University, Nashville, TN

Independent study conducted at Pace University (Pforzheimer Honors College), New York, NY.

Paper in press –
University of Michigan Undergraduate Journal of Public Health

BACKGROUND

8.3% **of U.S. adults** (≈21 million) had a major depressive episode in 2021 (NIMH)

280M **people worldwide** live with depression (WHO)

17.2% **of U.S. adults aged 18–25** had past-year depression in 2020, up from 10.3% in 2015 (Goodwin et al., 2022)

Treatment barriers
- High costs
- Long waitlists
- Provider shortages

Physical activity
- Low-cost
- No prescription
- Flexible, anywhere

Prior evidence

Physical activity: **17% lower odds** of developing depression (Kandola et al., 2019)

Half the recommended activity: **18% lower risk**; full amount: **25%** (Pearce et al., 2022)

Research gap: few nationally representative studies compare how often people exercise across several dimensions of depression.

AIMS

Question: Is how often U.S. adults do moderate physical activity linked to their self-reported depression and general health?

Hypothesis: More frequent moderate activity is associated with better depression-related outcomes.

Moderate physical activity
None · Low · Medium · High

linked to?

- General health
- Depression frequency
- Depression level

Controls: demographic & socioeconomic

METHODS

1 Data

IPUMS NHIS 2010–2023 · N = 332,521 U.S. adults · self-reported, nationally representative

2 Exposure: moderate activity, times/week

None 0 | Low 1–3 | Medium 4–6 | High 7+

3 Outcomes (3 separate models)

General health | Depression frequency | Depression level

4 OLS regression vs. no activity

(1) Activity only | (2) + Demographic | **(3) + Socioeconomic**

★ *Specification (3), with all controls, is preferred.*

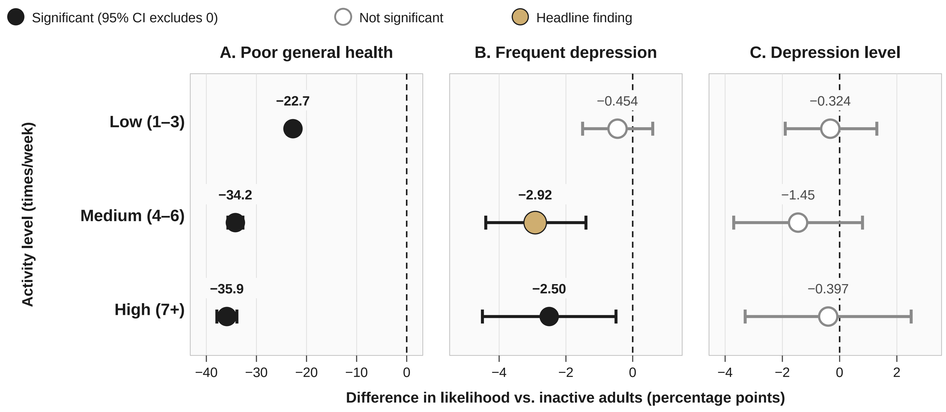
RESULTS

−2.92
percentage points

Adults active 4–6 times a week were 2.92 percentage points less likely to report frequent depressive symptoms than inactive adults.

95% CI: −4.4 to −1.4 · Specification 3

Figure 1. Change in each outcome vs. no activity, by activity level (Spec 3)



Dots = estimates; bars = 95% confidence intervals. Values below 0 mean less likely than adults with no moderate activity. Each panel has its own scale.

Figure 2. Sample size by moderate activity level (N = 332,521; % of sample)

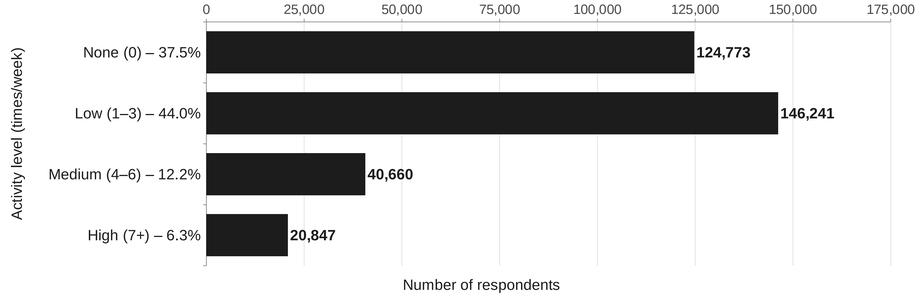


Table 1. Specification 3 estimates, percentage points [95% CI]

Activity level	General health (poor)	Depression frequency	Depression level
Low (1–3)	**−22.7*** [−23.8, −21.6]	−0.454 [−1.5, 0.6]	−0.324 [−1.9, 1.3]
Medium (4–6)	**−34.2*** [−35.7, −32.7]	**−2.92*** [−4.4, −1.4]	−1.45 [−3.7, 0.8]
High (7+)	**−35.9*** [−37.9, −33.9]	**−2.50*** [−4.5, −0.5]	−0.397 [−3.3, 2.5]
Observations	161,344	160,974	66,089
R^2	0.204	0.252	0.038

** Statistically significant (95% CI excludes 0). Reference group: no moderate activity.*

DISCUSSION

Modest but meaningful: 4–6 sessions a week was linked to fewer frequent depressive symptoms

Better self-rated health at every activity level (−22.7 to −35.9 pp)

Fits prior work linking activity to lower depression risk (Kandola et al., 2019; Pearce et al., 2022)

Plausible pathways: less stress, better sleep, greater well-being

No link with depression level: all CIs include 0 (R^2 = 0.038)

Socioeconomic status matters: welfare receipt consistently predicted worse outcomes (Alzahrani et al., 2024)

Limitations
- Self-reported data: social desirability & extreme response bias
- Unobserved confounders (e.g., sleep, genetics)
- Cross-sectional data show association, not cause
- Income measured only in categories

CONCLUSIONS

1. **Being moderately active 4–6 times a week was linked to a lower chance of feeling depressed often.**
2. Any regular moderate activity – even 1–3 times a week – was linked to better self-rated health.
3. Exercise is not a cure: it was not linked to how severe depression feels, so it should complement, not replace, treatment.

Future directions
- Test activity as a treatment for people clinically diagnosed with depression
- Track real-time, individual-level improvement

ACKNOWLEDGEMENTS

I thank Professor Osborne for his helpful feedback on the manuscript, and my mom and dad for their constant support and encouragement.

Funding: This independent study received no external funding.

Data: IPUMS NHIS v8.1 (Blewett et al., 2025).

Author Contribution Statement
NG (sole author) conceived and designed the study, prepared the data, performed all analyses, interpreted the results, wrote the manuscript, and designed this poster.

Selected references

Blewett, L. A., et al. (2025). IPUMS Health Surveys: NHIS (Version 8.1). doi.org/10.18128/D071.V8.1
Goodwin, R. D., et al. (2022). Am. J. Prev. Med., 63(5), 726–733.
Kandola, A., et al. (2019). Neurosci. Biobehav. Rev., 107, 525–539.
Pearce, M., et al. (2022). JAMA Psychiatry, 79(6), 550–559.
Alzahrani, H., et al. (2024). J. Adv. Trends Med. Res., 862–869.
NIMH (n.d.). Major Depression. · WHO (2025). Depressive disorder (depression).